

Khai-Nguyen Nguyen

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RESEARCH INTERESTS

I am interested in multimodality, robustness, AI safety and their applications. To this end, I have worked on: **data** [C1][C5][C7], **evaluation** [C2][C3], **training** [C1][C5], and **post-hoc methods** [C4][C8].

EDUCATION

University of Virginia 2026 – 2029 (expected)
Ph.D. student in Data Science Advisor: [Chirag Agarwal](#)

College of William and Mary 2023 – 2026
M.S. student in Computer Science Advisor: [Antonio Mastropaolo](#)
GPA: 3.96/4.00

Bucknell University 2019 – 2023
B.Sc. in Computer Science and Engineering; minor in Statistics
GPA: 3.94/4.00; *summa cum laude*

RELEVANT EXPERIENCE

Nguyen Lab, Auburn University Mar 2025 – May 2026
Remote Collaborator Advisor: [Anh Totti Nguyen](#)

- Led research on knowledge-prior bias in LLMs: (1) constructed an evaluation benchmark, (2) analyzed model behavior using mechanistic interpretability techniques (e.g., activation patching and probing), and (3) developed a distributed multi-GPU training pipeline based on [OLMo-core](#) to mitigate bias. This work resulted in co-first-authored paper [C3] and co-authored paper [C1].

CodaMetrix, Boston May – Aug 2024, 2025
Machine Learning Intern Advisor: [Cheng Li](#)

- Developed a Gemini-based judge to evaluate and refine medical entity extractions from health records, improving overall F1 by 3% over the baseline.
- Designed & deployed a medical entity extraction system on Databricks with DSPy prompt optimization.
- Proposed and trained a BERT-based model for CodaMetrix's multilabel ICD-10 classification system on 1M samples, improving accuracy by 4% over the deployed system.

FPT Software AI Center, Vietnam Jan – Sep 2024
Remote Research Intern Advisor: [Truong-Son Hy](#)

- Led research on trustworthy LLMs for healthcare applications: (1) constructed large-scale multimodal training datasets with 25k–30k samples and (2) developed CoT distillation pipelines. This work resulted in co-first-authored papers [C5][C7], and co-authored paper [C6].

SELECTED PUBLICATIONS

* Equal contribution.

- [C1] **Deep and shallow biases in language models** [\[Paper\]](#)
An Vo, Vy Tuong Dang, **Khai-Nguyen Nguyen**, Emilio Villa-Cueva, Tamar Solorio, Anh Totti Nguyen, Daeyoung Kim
Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)
- [C2] **Pattern over Pixels: Measuring Pattern Completion Bias in Multimodal Code Generation** [\[Paper\]](#)
Khai-Nguyen Nguyen, Oscar Chaparro, Antonio Mastropaolo
IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)
- [C3] **Vision Language Models are Biased** [\[Paper\]](#)
An Vo*, **Khai-Nguyen Nguyen***, Mohammad Reza Taesiri, Vy Tuong Dang, Anh Totti Nguyen, Daeyoung Kim
International Conference on Learning Representation (ICLR 2026)

- [C4] **Toward Explaining Large Language Models in Software Engineering Tasks** [\[Paper\]](#)
Antonio Vitale*, **Khai-Nguyen Nguyen***, Denys Poshyvanyk, Rocco Oliveto, Simone Scalabrino, Antonio Mastropaolo
ACM Transactions on Software Engineering and Methodology (TOSEM 2026)
- [C5] **Sentiment Reasoning for Healthcare** [\[Paper\]](#)
Khai-Nguyen Nguyen*, Khai Le-Duc*, Bach Phan Tat, Duy Le, Long Vo-Dang, Truong-Son Hy
Annual Meeting of the Association for Computational Linguistics (ACL 2025, Industry Track, Oral)
- [C6] **Medical Spoken Named Entity Recognition** [\[Paper\]](#)
Khai Le-Duc, David Thulke, Hung-Phong Tran, Long Vo-Dang, **Khai-Nguyen Nguyen**, Truong-Son Hy, Ralf Schlüter
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025, Industry Track)
- [C7] **Real-time Speech Summarization for Medical Conversations** [\[Paper\]](#)
Khai Le-Duc*, **Khai-Nguyen Nguyen***, Long Vo-Dang, Truong-Son Hy
Conference of the International Speech Communication Association (Interspeech 2024, Oral)
- [C8] **Getting Away with More Network Pruning: From Sparsity to Geometry and Linear Regions** [\[Paper\]](#)
Jeffrey Cai*, **Khai-Nguyen Nguyen***, Nishant Shrestha, Aidan Good, Ruisen Tu, Xin Yu, Shandian Zhe, Thiago Serra
International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR 2023); Sparsity in Neural Networks Workshop at ICLR 2023

PREPRINTS

- [P1] **Prior Bias in Vision Language Models on UML Diagram Interpretation** [\[Paper\]](#)
Zaiyu Cheng*, **Khai-Nguyen Nguyen***, Antonio Mastropaolo
arXiv preprint, 2026
- [P2] **S-Chain: Structured Visual Chain-of-Thought for Medicine** [\[Paper\]](#)
Khai Le-Duc et al.
arXiv preprint, 2025

AWARDS & HONORS

Computer Science Fellowship , College of William and Mary	2023
Dean's List , Bucknell University (every semester)	2019 – 2023
Program for Undergraduate Research Grant , Bucknell University (\$4,000)	2021
Honorable Mention , Mathematical Contest in Modeling	2020
Awarded Honorable Mention out of 13,753 teams worldwide (top 26%).	

INDUSTRY & PRESS COVERAGE

Industry Usage

2026: [Qwen used VLMsAreBiased to evaluate Qwen3.8-Max.](#)

2025: [Google DeepMind used VLMsAreBiased to evaluate Gemini 3 Pro.](#)

News Coverage

2026: Featured on Sky News' [video](#) and [article](#) on potential dangers of AI in modern warfare.

TEACHING & SERVICES

Teaching Assistant, University of Virginia: Computation for Data Science.

Teaching Assistant, William and Mary: Intro to Data Science; Computer Organization; Software Engineering.

Teaching Assistant, Bucknell University: Data Structures and Algorithms; Calculus I and II.

Reviewer: ACL Rolling Review (ARR), May 2026.